Asteroid Tracking & Intercept with Convex Optimization and State Estimation Techniques

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***Abstract*—In this paper, we address the challenges of state estimation in the nonlinear and dynamic context of intercepting an asteroid with a spacecraft. Given the inherent complexities of the environment and difficult to model deterministic perturbations like solar radiation, atmospheric drag, and gravitational anomalies, traditional deterministic modeling can fail to capture spacecraft dynamics adequately. Therefore, robust state estimation techniques are required to properly localize spacecrafts. We use a combination of Kalman filters—specifically the extended Kalman filter (EKF), the unscented Kalman filter (UKF), and the particle filter—to estimate the states of both a spacecraft and an asteroid during a 2D intercept mission. We integrate simplified dynamic models with these filters to evaluate their efficacy under the influence of random perturbations and sensor error. Furthermore, we employ convex optimization for feedback control to guide the spacecraft along a desired trajectory. With each of the filters' state estimates, the feedback controller was able to keep the spacecraft on the planned trajectory and reach the desired intercept location. However, only the particle filter was able to effectively estimate both the spacecraft and the asteroid's state. Both the EKF and UKF fail to produce accurate estimates of the asteroid's state during the simulation.**

I. Introduction

Spacecraft dynamics, though largely deterministic, are highly nonlinear. The effects of perturbations such as solar radiation pressure, atmospheric drag, and gravitational irregularities are also deterministic and can be represented in a nonlinear, closed form. However, due to numerous nonlinear factors that go into deterministic spacecraft dynamics, it becomes difficult to model and can be desirable to instead model dynamics probabilistically, with quantifiable uncertainty. In this project, we aim to maintain accurate estimates of both a spacecraft's and an asteroid's state during an intercept trajectory. To do this, we will use simplified dynamics models in conjunction with various filters to maintain some degree of confidence in time-dependent states without analytically considering the complex effects of perturbations.

II. Problem Statement

Our problem statement is twofold. Of the extended Kalman filter (EKF), unscented Kalman filter (UKF), and particle filter, which performs the best with the nonlinear spacecraft dynamics and can best estimate the asteroid's state? Secondly, can we effectively remain on a desired trajectory using these estimated states and a convex optimization based feedback controller?

III. Related Work

A. Sources [1] [2] [9]

These papers each give methods of formulating the dynamical equations for a rocket. [2] develops a full 6DOF 3D rocket dynamics model while incorporating all forms of disturbance in the model. The complexity of this model is high, but it is also the most realistic model. While [1] and [9] derive models for a "restricted" rocket model that only has 3DOF. For our project a model more similar to theses "restricted" rockets is enticing due to the time constraints. [1] argues that for rockets, the rotational motion is less important than the translational when generating a trajectory; therefore we can still generate accurate trajectories even when neglecting these additional rotational possibilities.

B. Source [5]

This paper provides a motivation for state estimation in the vicinity of an asteroid. Because asteroids can have highly irregular gravitational fields - rotating ones at that - the predictability of spacecraft dynamics becomes washed away in a sea of perturbations, necessitating active state estimation.

C. Source [6]

This suggests that using a coordinate frame centered about the spacecraft (the Hill frame) simplifies measurement computation. This paper also considers the effects of J2 perturbations and solar radiation pressure to make their simulations more accurate. In our project, we can stand to benefit by excluding these considerations in our model, thus learning how various filters adapt to these perturbations.

D. Source [7]

Because there are certain model parameters that cannot be known (for instance, the asteroid's mass distribution), this paper suggests using model-adaptive control to iteratively find the parameters which best fit observed measurements. In practice, this is as simple as adding these unknown parameters to the dynamic state and measurement/transition models.

E. Source [10] [11]

Finally, [10] details the typical sensor suite used on a rocket. The rocket described in this paper uses a combination of an IMU to obtain the angular rates and translational accelerations and an actuator encoder to determine the nozzle angle. This is a similar suite that we can implement on our project. Additionally, [11] suggests how to implement an extended Kalman filter and an unscented Kalman filter for the purposes of determining a rocket's attitude. In our project, we can implement an augmented form of these filters as we will be using a restricted form of the attitude and also include the translational position.

IV. Problem Formulation

A. Problem Assumptions

For this project, we will assume that the asteroid is under the influence of the Earth's gravity using a spherical Earth model, but the spacecraft is not. Additionally, we will assume that we have an initial estimate of the spacecraft and asteroid's state with some covariance. Our initial confidence in our spacecraft's state will be higher than the initial confidence in the asteroid and that will be reflected in the covariance matrices. Lastly, we will assume that the feedback controller is designed to follow a trajectory that takes a straight line path from the spacecraft's initial position to the ideal final position of the asteroid.

B. System Modelling

Taking into account the considerations above, the feedback system can be modelled as 4 subsystems: the spacecraft-asteroid dynamics models, the measurement model, the state estimator, and the controller.

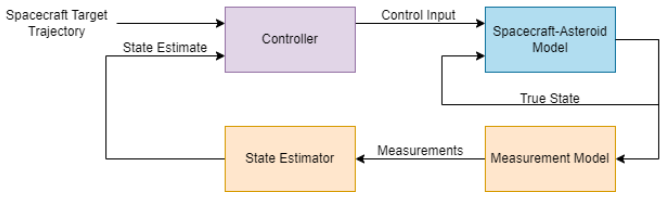


Fig. 1. System block diagram

After modelling the spacecraft-asteroid system's dynamics and the measurement model, the controller and state estimator can be designed independently and integrated as shown in figure 1. Therefore, we will begin by modelling the dynamics model and measurement model.

C. Dynamics and Measurement Models

We can define the dynamics of the asteroid and the spacecraft separately in our system since the dynamics of the two objects do not couple. Therefore, we will begin with defining the spacecraft dynamics.

1) Spacecraft Dynamics: As stated previously, for simplicity, we will be focusing on a 2D planar formulation of a spacecraft. The planar spacecraft has 3 degrees of freedom: the x position, the y position, and the heading θ. With these degrees of freedom, we can define a state vector for the spacecraft, $s_{sc} = [x, y, \theta, v, \omega, a]$. We define the remaining states such that v represents the spacecraft's translational speed, ω represents the heading's angular velocity, and a represents the translational acceleration.

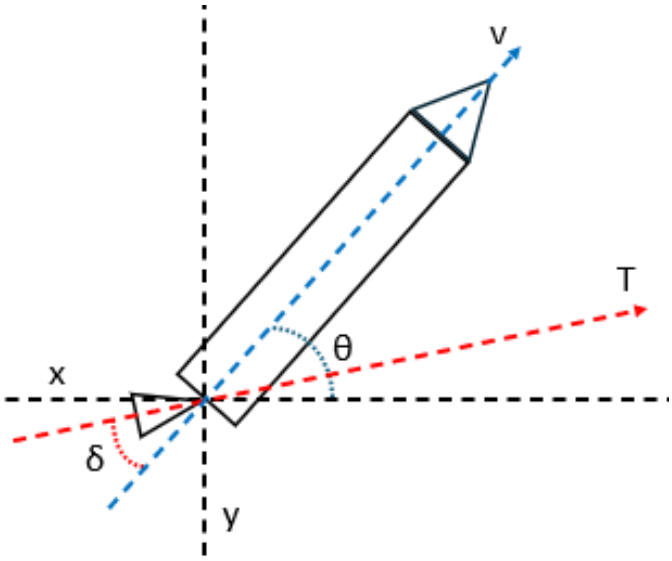


Fig. 2. Spacecraft model

To control the spacecraft, we implement a thrust vector control scheme that determines a thrust magnitude T and a nozzle angle δ (this angle is with respect to the heading angle). The projection of the thrust direction onto the heading direction is the thrust's contribution to the translational acceleration. The portion of the thrust orthogonal to the heading direction is the thrust's contribution to the angular acceleration of the spacecraft. With this control system, we can define the spacecraft's state dynamics as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{v} \\ \dot{\omega} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} v\cos(\theta) \\ v\sin(\theta) \\ \omega \\ a + \frac{T}{m}\cos(\delta) \\ \frac{T}{ml^2}\sin(\delta) \\ 0 \end{bmatrix} \tag{1}$$

where m and l are the mass of the spacecraft and distance to the center of mass respectively. For simplicity, we assume the spacecraft's mass remains constant, which also implies a constant distance to the center of mass. These non-linear dynamics pose a challenge for the proposed convex optimization-based controller and the extended Kalman filter. Therefore, we can perform a linearization of the dynamics by solving for the Jacobians of the dynamics with respect to the states and the control inputs. First, we can take the partial derivatives of

the dynamics with respect to the states to generate the state dynamics Jacobian matrix A_{sc}:

$$A_{sc} = \begin{bmatrix} 0 & 0 & -v\sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & v\cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{2}$$

Next, we can take the partial derivatives of the dynamics with respect to the control inputs to generate the control input Jacobian B_{sc}:

$$B_{sc} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m}\cos(\delta) & -\frac{T}{m}\sin(\delta) \\ \frac{1}{ml^2}\sin(\delta) & \frac{T}{ml^2}\cos(\delta) \\ 0 & 0 \end{bmatrix} \tag{3}$$

2) Asteroid Dynamics: To match the planar formulation of the spacecraft, we will assume that the asteroid and spacecraft exist within the same plane. Therefore, we can define the asteroid using a 2D reduction of the fundamental orbital equation. With this equation, we will define the asteroid's state vector as $s_{ast} = [x, y, v_x, v_y]$ and the corresponding dynamical equations as:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{v}_x \\ \dot{v}_y \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ \frac{-\mu x}{(x^2+y^2)^{\frac{3}{2}}} \\ \frac{-\mu y}{(x^2+y^2)^{\frac{3}{2}}} \end{bmatrix} \tag{4}$$

In these equations, μ represents Earth's gravitational constant. Unlike the spacecraft system, the asteroid represents an autonomous dynamical system; therefore, there is no input. Once again, these dynamics are non-linear so we must linearize them for the convex optimizer and the extended Kalman filter. Therefore, we will linearize in the same way as the spacecraft's dynamics. Given that there are no inputs to this system, we will only solve for the state dynamics Jacobian matrix A_{ast}:

$$A_{ast} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{2(x^2-y^2)\mu}{(x^2+y^2)^{\frac{5}{2}}} & \frac{3\mu xy}{(x^2+y^2)^{\frac{5}{2}}} & 0 & 0 \\ \frac{3\mu xy}{(x^2+y^2)^{\frac{5}{2}}} & \frac{2(y^2-x^2)\mu}{(x^2+y^2)^{\frac{5}{2}}} & 0 & 0 \end{bmatrix} \tag{5}$$

3) Joint Dynamics: We can combine the two systems simply by combining the individual system states as $s = [s_{\text{sc}}, s_{\text{ast}}]$. The spacecraft and asteroid systems have no influence on each other, and as such there is no coupling in the dynamics of the two systems. Therefore, the state dynamical equations are simply a concatenation of each individual system. This gives the following combined state dynamics Jacobian and control input Jacobian:

$$\tilde{A} = \begin{bmatrix} A_{sc} & 0 \\ 0 & A_{ast} \end{bmatrix} \quad \tilde{B} = \begin{bmatrix} B_{sc} \\ 0 \end{bmatrix} \tag{6}$$

To make the convex optimization tractable, we will use a discretized form of the dynamics through the forward Euler integration method:

$$\begin{aligned} s_{t+1} &= s_t + \dot{s}_t \Delta t \\ &= (I + \tilde{A}\Delta t)s_t + \tilde{B}\Delta t u_t \\ &= As_t + Bu_t \end{aligned} \tag{7}$$

4) Measurement Model: The spacecraft will use a combination of ranging, IMU, and a laser rangefinder to estimate the system states. The ranging system gives a measurement of the spacecraft's position in inertial space, the IMU gives a measurement of the spacecraft's angular velocity and acceleration, and the laser rangefinder gives a measurement of the asteroid's distance relative to the spacecraft. We can define the measurement vector as:

$$y = \begin{bmatrix} x^{\text{sc}} \\ y^{\text{sc}} \\ \omega^{\text{sc}} \\ a^{\text{sc}} \\ \left|\left| \begin{bmatrix} x^{\text{ast}} \\ y^{\text{ast}} \end{bmatrix} - \begin{bmatrix} x^{\text{sc}} \\ y^{\text{sc}} \end{bmatrix} \right|\right|_2 \end{bmatrix} \tag{8}$$

Because, this measurement vector is non-linear, we will need to linearize the measurement for the extended Kalman filter. For the measurement vector, we will compute the Jacobian by taking the partial derivative with respect to both the spacecraft's and asteroid's states. Therefore, we get the following measurement Jacobian matrix C:

$$\begin{aligned} d_x &= (x^{\text{ast}} - x^{\text{sc}}) \\ d_y &= (y^{\text{ast}} - y^{\text{sc}}) \\ d &= \sqrt{d_x^2 + d_y^2} \\ C &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{-d_x}{d} & \frac{-d_y}{d} & 0 & 0 & 0 & 0 & \frac{d_x}{d} & \frac{d_y}{d} & 0 & 0 \end{bmatrix} \end{aligned} \tag{9}$$

D. Controller Design

Now that the dynamics and measurement models of the spacecraft-asteroid system have been defined, the next step is to define the controller. To control the spacecraft, we are using a model predictive control scheme with convex optimization. Model predictive control is a form of finite time horizon optimal control defined as:

$$\begin{aligned} \min_{u_{1:N-1}} \quad & f_0(s,u) \\ \text{subject to } & s_0 = \hat{s}_0 \\ & s_{t+1} = As_t + Bu_t \ \forall\ t \in \{0,\dots,N-1\} \\ & u_t \in \mathcal{U} \ \forall\ t \in \{0,\dots,N-1\} \end{aligned} \tag{10}$$

where $\hat{s}_0$ is the initial state provided by the state estimator, u is the set of control inputs, and $\mathcal{U}$ is the set of feasible control inputs. To apply convex optimization techniques to this problem, the objective function must be convex and the constraining set must be convex sets. Given that the dynamics are inherently non-convex, we can use the linearized discrete dynamics of the combined spacecraft-asteroid system defined in the previous section for the dynamic constraints in the MPC problem. Finally, we can define input constraint set as the following set:

$$\begin{aligned} & U_{\text{low}} = \begin{bmatrix} 0 \\ \delta_{\text{low}} \end{bmatrix} \ U_{\text{high}} = \begin{bmatrix} T_{\text{high}} \\ \delta_{\text{high}} \end{bmatrix} \ U_{\text{smooth}} = \begin{bmatrix} T_{\text{smooth}} \\ \delta_{\text{smooth}} \end{bmatrix} \\ & \mathcal{U} = \{u_t \mid U_{\text{low}} \preceq u_t \preceq U_{\text{high}}, |u_t - u_{t-1}| \preceq U_{\text{smooth}}\} \end{aligned} \tag{11}$$

where U_{low} and U_{high} are the lower and upper thresholds on the control inputs and U_{smooth} is a smoothness constraint that prevents large jumps between consecutive control inputs. Next, we must define the objective function. For the MPC to act as a proper feedback controller, it should track a desired state trajectory at all times. Therefore, we can define the objective as minimizing the sum of the squared l_2 distance norms between the spacecrafts current state and the desired state over the full horizon.

$$f_0(s) = \sum_{i=0}^{N} ||s_t - \tilde{s}_t||_2^2 \tag{12}$$

This generates the following optimization problem:

$$\begin{aligned} \min_{u_{1:N-1}} \quad & \sum_{i=0}^{N} ||s_t - \tilde{s}_t||_2^2 \\ \text{subject to } & s_0 = \hat{s}_0 \\ & s_{t+1} = As_t + Bu_t \ \forall\ t \in \{0,\dots,N-1\} \\ & u_t \in \mathcal{U} \ \forall\ t \in \{0,\dots,N-1\} \end{aligned} \tag{13}$$

E. State Estimators

As stated, for this project we will use three different state estimators, an extended Kalman filter, an unscented Kalman filter, and a particle filter. Each filter will be used online with the feedback controller; therefore, the state estimate and covariance will be continuously updated as we take actions and measurements. Each filter will include a process noise estimate Q and a measurement noise estimate R. For fair comparison of each filter, they will all use the same noise estimates.

V. Results

A. Setup

To test each filters' ability to estimate the spacecraft's and asteroid's states, we will design a test where the asteroid is in some orbit around the Earth, and the spacecraft is in the same orbit but at a smaller initial true anomaly. We will begin with designing the initial conditions on the asteroid. Due to the large distance values that occur in the space environment, we will use canonical units for our initial conditions. The conversion from canonical units based on the Earth to SI units is describe in table I.

Quantity	Canonical Units	SI Units
Distance	1 DU	6378.136 km
Time	1 TU	806.8118744 s
Velocity	1 $\frac{\text{DU}}{\text{TU}}$	7.9053628 $\frac{\text{km}}{\text{s}}$

TABLE I
CANONICAL UNITS TO SI UNITS CONVERSION

Using canonical units, we define the asteroid's initial conditions as:

$$s_0^{\text{ast}} = \begin{bmatrix} 1.49999 \\ 0.004712 \\ -0.002565 \\ 0.816493 \end{bmatrix} \tag{14}$$

Next, we define the initial conditions of the spacecraft to be positioned in the same orbit, but at a slightly smaller true anomaly. The spacecraft is then oriented such that it is parallel with the straight line trajectory to the desired intercept point with the asteroid. Therefore, we get the following initial conditions of the spacecraft as:

$$s_0^{\text{sc}} = \begin{bmatrix} 1.5 \\ 0.0 \\ 1.62047 \\ 1.05519 \\ 0.0 \\ 0.0 \end{bmatrix} \tag{15}$$

Because we have monitored the spacecraft from launch to our current time, we have a much higher level of confidence in the state of our spacecraft than we do the asteroid. Therefore, we will set the prior covariance of the spacecraft to $\Sigma_0^{\text{sc}} = 0.001I$ and we will set the prior covariance of the asteroid to $\Sigma_0^{\text{ast}} = 0.04I$. Therefore, the initial estimate and covariance for our filters will be:

$$\begin{aligned} \mu_0 &= \begin{bmatrix} s_0^{\text{sc}} & s_0^{\text{ast}} \end{bmatrix} \\ \Sigma_0 &= \begin{bmatrix} \Sigma_0^{\text{sc}} & 0 \\ 0 & \Sigma_0^{\text{ast}} \end{bmatrix} \end{aligned} \tag{16}$$

As stated in the problem formulation, for a fair comparison we will use the same estimate of process and measurement noise for each filter. For this test the filter process and measurement noises are set such that $Q = R = 10^{-5}I$. Once the initial conditions are set, we simulate actions from the feedback

controller using the true (non-linear) dynamics. To model the possible perturbations in the environment, white Gaussian noise (with variance 10^{-5}) is added to the spacecraft's change in acceleration and angular velocity and the asteroid's change in velocity at each simulation step. Additionally, to model the imperfect sensor measurements, zero-mean white Gaussian noise is added to each of the measurement outputs with covariance relative to the accuracy of the sensor. Table II lists the variances for the sensors used in this test.

Sensor	Measurement Variance
Ranging	10^{-5}
IMU	10^{-5}
Laser Rangefinder	10^{-3}

TABLE II
SENSOR VARIANCE USED IN TEST

For this test, the simulation lasts for 950 times steps with a step time of $\Delta t = 0.0806$. When analyzing each of the filters in this test, we will overlay the true state and the filter's belief of the state for both the spacecraft and asteroid. The target position will also be include to show that the feedback control is able to stay on course using the provided state estimate. Finally, the absolute value of the error between the estimate and the true state will be compared for each filter to determine which has the best tracking ability.

B. Extended Kalman Filter

The first filter tested was the extended Kalman filter (EKF). This filter utilizes the linearized model of the spacecraft-asteroid dynamics and the linearized measurement model derived above. Of the three filters tested, the EKF is the most basic. It is able to track the spacecraft's state relatively well as shown in figure 3, and the spacecraft is able to stay on the target path with the estimates obtained from this filter. However, when accounting for the y-scaling of each plot, the error in the x position of the spacecraft is much more significant than the other states.

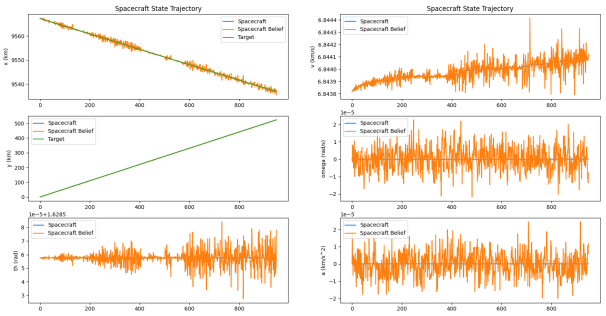


Fig. 3. Results of the EKF on tracking the spacecraft state.

Upon reviewing the filter's ability to track the asteroid state we find that the EKF runs into significant errors when estimating the x and y positions of the asteroid. Given the y-scaling of the plots, the estimates are approximately 30 km off on both position values. This is likely because the only measurement of the asteroid's position we have is relative to the spacecraft's position. Therefore the noise that we see in the spacecraft's position measurement affects our estimation of the asteroid's position.

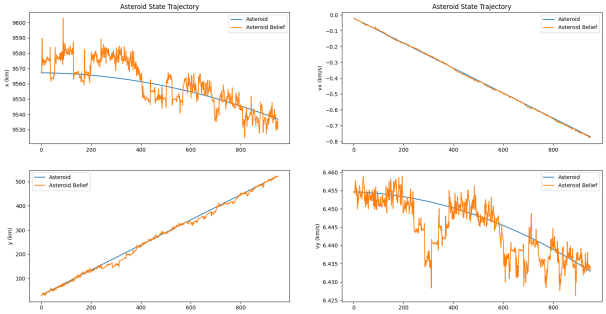


Fig. 4. Results of the EKF on tracking the asteroid state.

C. Unscented Kalman Filter

The second filter that was tested was the unscented Kalman filter (UKF). Unlike the EKF, the UKF re-parameterizes a Gaussian to the form of $2n + 1$ sigma points where n is the dimensionality of the system's state. For this project, we will have 21 sigma points and a scaling $\lambda = 2$ in the unscented transform. This is meant to address the errors that arise from linearizing the dynamics for propagating the posterior mean and covariance in the EKF. Because the UKF only uses the true dynamics and measurement models, we expected the UKF to perform the best, but this was not necessarily the case. When comparing the spacecraft state belief to the true state, the UKF performed very well for the states that we directly measure (x, y, ω, and a). For the states that we don't directly measure, we see errors of a much higher magnitude and the estimated and true state seem to diverge. Even with these diverging states, the feedback controller is still able to use the estimate to track the ideal position path very well. This is likely due to the small time frame of the simulation and if it were to continue for longer the spacecraft would likely diverge from the target path.

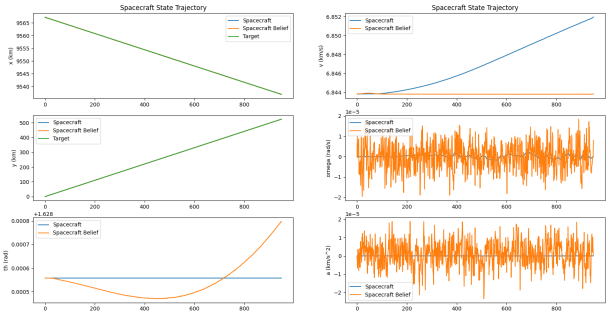


Fig. 5. Results of the UKF on tracking the spacecraft state.

When analyzing the UKF's ability to track the asteroid state, the performance degrades immensely. While the filter tracks the asteroid's x-velocity rather well, we see large discrepancies in the x and y positions and the y velocity. The filter's belief of the x and y position of the asteroid seems to be identical to it's belief of the spacecraft's position. Because the velocity of the asteroid is a function of the position and there are no measurements of the velocity, the large initial errors in the y position likely cause the divergence seen in the y velocity. Overall, while the UKF has good performance for the spacecraft states, it completely fails to estimate the asteroid's state.

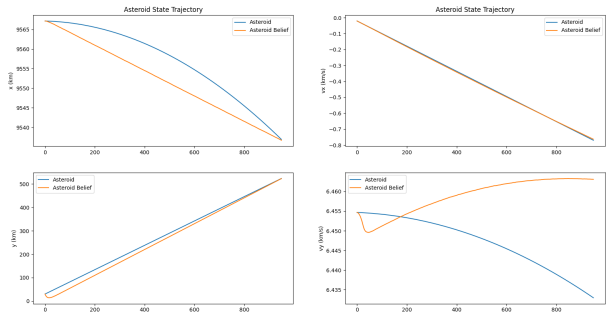


Fig. 6. Results of the UKF on tracking the asteroid state.

D. Particle Filter

The final filter tested was the particle filter. When running simulations with a particle filter, it is recommended to use approximately 10^n particles to represent the posterior state estimate. Our joint spacecraft-asteroid state is 10-dimensional, meaning that we would need to use ten billion particles for an accurate estimate. Of course, for computational reasons, this is impractical. Therefore, we ultimately ran our filter with 10^4 particles. Even with the suboptimal number of particles, the particle filter obtains great results initially when tracking the spacecraft; however, we do begin to see some divergence from the true state as the simulation ends. However, even with this divergence, the feedback controller is still able to keep the spacecraft on the ideal position path as shown by the x and y true states, target state, and believed state all aligning.

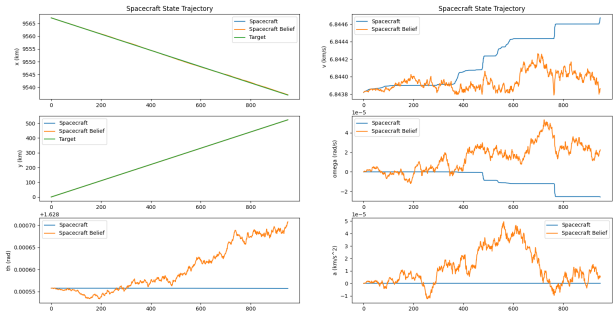


Fig. 7. Results of the Particle filter on tracking the spacecraft state.

Interestingly, unlike the EKF and UKF, the particle filter's estimate of the asteroid state almost perfectly tracks both the position and velocity.

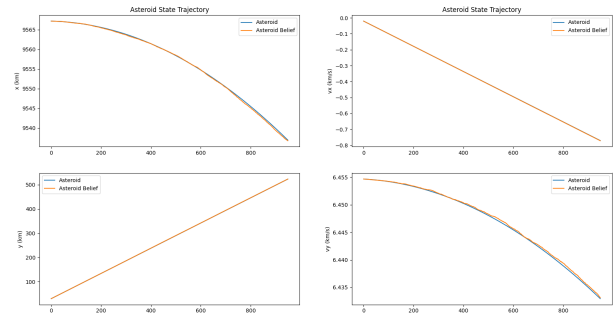


Fig. 8. Results of the Particle filter on tracking the asteroid state

E. Comparison

After testing all three filters, we can compare the estimate errors for each filter. Beginning with each filter's ability to track the spacecraft's state error, when accounting for the y-scaling of each state's plot, the UKF and particle filter outperform the EKF. We are able to track the spacecraft's position within about 50-100 meters with both the UKF and particle filter, while the EKF performs noticeably worse with errors of up to 2 km. Because of the increase in error exhibited by the UKF in the heading angle and velocity, the particle filter has the most consistent performance across the states out of the three filters.

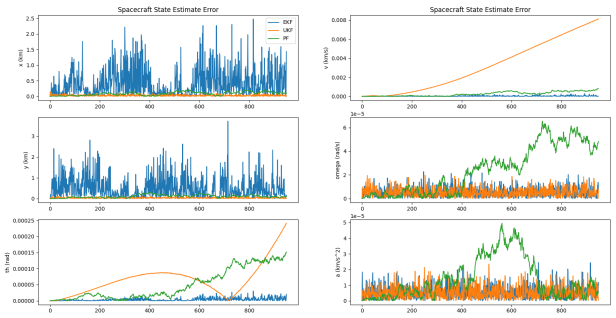


Fig. 9. Spacecraft state estimate error comparison between all 3 filters

When comparing each filters' ability to estimate the asteroid's state, the UKF and EKF both perform significantly worse than the particle filter. The UKF and EKF get errors in the asteroid's x and y position estimate upwards of 30 and 40 km and errors. Out of the 3 filters, only the particle filter can accurately track the asteroid's state.

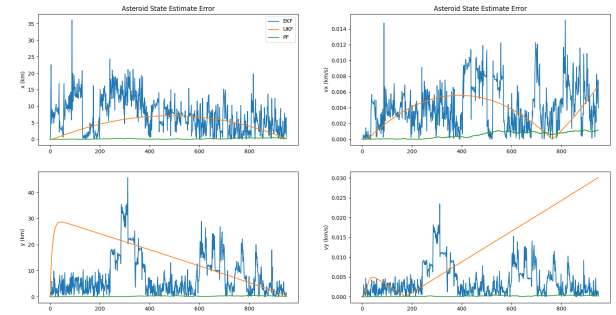


Fig. 10. Asteroid state estimate error comparison between all 3 filters

Given the results above, when considering performance on estimating both the asteroid's state and the spacecraft's state, the particle filter is by far the best filter for this system. However, there is a significant drawback to using the particle filter due to the time it takes to estimate the state. Table III lists the average time it takes for each filter to generate a new estimate. The particle filter takes ≈ 0.3 seconds which is ≈ 50 times longer than the UKF and ≈ 1250 times longer than the EKF which significantly hampers the viability of this system.

Filter	Average Time Per Estimate (s)
EKF	0.0002669486828241602
UKF	0.0068768481559538484
PF	0.33422602718651284

TABLE III
AVERAGE FILTER ESTIMATE TIME

Overall, the particle filter still remains the best filter for the system given the significant performance advantage it offers and can be sped up through parallel programming.

VI. Limitations

While the designed system is able to track the spacecraft and asteroid's states well when using the particle filter, there are some limitations on the design. The current control design is only able to track a pre-planned trajectory and is unable to path plan a new one. This is because the MPC controller is based on a convex optimizer which takes in a linearized form of the dynamics. This linearization is based on a first order Taylor expansion which is only accurate close to the linearization point. Therefore, when path planning for distances far from the initial location, the controller struggles to find a path as the dynamics its optimizing over are only accurate near the initial condition. Additionally, the simulation models used for the spacecraft and asteroid dynamics are simplified and may not fully capture the real-world physics, particularly in the non-linear regimes where model inaccuracies become more pronounced. This simplification affects the fidelity of the state estimation and the effectiveness of the control inputs computed by the optimizer. Another significant limitation is the computational demand of the state estimators, particularly the particle filter, which requires a large number of particles to adequately represent the system's state space. This computational burden can limit the practical deployment of such filters in real-time scenarios, where rapid state estimation is crucial.

Furthermore, the environmental modeling within the simulation lacks certain real-world perturbations such as the Yarkovsky effect and non-gravitational forces, which can influence the trajectory of the asteroid significantly over time. The absence of these factors could lead to deviations between the simulated and actual missions, reducing the accuracy of our predictions and the robustness of the control strategy.

Finally, the reliance on specific initial conditions for system performance indicates a potential fragility in handling unexpected scenarios or errors in initial state estimation. This could be problematic in actual space missions where initial conditions cannot always be precisely controlled or predicted.

VII. Conclusions

In conclusion, the particle filter is the only filter to perform well when estimating the asteroid's state and the spacecraft's state. The UKF and EKF each performed well when estimating many of the spacecraft state but performed abysmally when estimating the asteroid's state. Therefore, we conclude that the particle filter would be the most robust in a real world asteroid-collision scenario and should be used in the loop to help path plan and guide a the spacecraft. However, due to the controller's inability to path plan, the designed controller is inadequate for this task. Moving forward, addressing the shortcomings of the current controller will be essential for enhancing mission flexibility and reliability. The introduction of an adaptive or learning-based control system could significantly improve the ability to respond to dynamic environmental conditions and unforeseen challenges during the mission. Additionally, further research into more comprehensive state

estimation techniques that can seamlessly integrate various sensor inputs and respond dynamically to changes in system behavior would be beneficial. These improvements are crucial for advancing the field of aerospace engineering, particularly for complex missions involving irregular celestial bodies and extended space travel.

VIII. Future Work

Moving forward, we would like to use our conclusions to create a more realistic controller that might one day be used on an asteroid intercept mission. With the limitations on the convex MPC's ability to path find, a new route could be to use a nonlinear optimizer to find a path and implement gain scheduled LQR. Additionally, we will cut down the project by only using the particle filter, and re-consider certain assumptions we made which may not be realistic. First, we would like to extend our project to 3 dimensions while ensuring feasible computation time. Next, we will include the gravitational affects of the earth so that our model is a better representation of the true underlying physics. Finally, we would like to replan the trajectory based on the spacecraft's estimate of it's own and the asteroid's state

IX. Contributions

For this project, Chris designed and coded the dynamic and measurement model for the spacecraft-asteroid system, helped design and code the model predictive controller, implemented the EKF, and integrated each portion of the system together. Ashok implemented the PF and tested different obejctive functions for the model predictive controller. Graham implemented the UKF, tested the fully integrated system, designed the asteroid orbit for the test scenario, and developed the animated plots used in the presentation.

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